Deep Residual Learning of a Probabilistic' Partial Least Squares Model for Predictive Data Analytics

Zhiqiang Ge, *Senior Member, IEEE*, Duxin Chen, *Member, IEEE*, and Wenwu Yu, *Senior Member, IEEE*

***Abstract*—Recently, probabilistic latent variable models have played an important role in data analytics in various industrial application scenarios, such as process monitoring, fault diagnosis, and soft sensing. Inspired by the idea of lightweight deep learning, this article proposes a new deep residual learning method for the probabilistic' partial least squares (PLSs) model. First, layerwise probabilistic modeling is carried out to extract supervised latent variables in different hidden layers of the deep model using a well-designed expectation-maximization algorithm for parameter optimization. Through this layerwise residual learning process, more target-related latent variables can be extracted, which are supervised by the outputs of the predictive model. Next, an additional probabilistic model is constructed for information fusion and further extraction of supervised latent variables which are highly related to the modeling target. In fact, this step can be considered as an ensemble learning strategy, which has great potentials in decreasing modeling error and reducing prediction uncertainty. A soft-sensing strategy is then developed for online prediction of key variables. The performance is evaluated using two industrial examples. Compared to the shallow probabilistic model, the performance of the deep model has been improved by 10%–20%.**

***Impact Statement*—PLSs model has been widely used for predictive data analytics. While a lightweight deep learning form of the basic model has recently been developed, it still lacks probabilistic interpretations for predictive data modeling and analysis. Inspired by the ideas of both deep learning and residual learning, a new deep probabilistic learning form of the PLSs model is proposed in this article. Through a layer-by-layer residual learning process, more target-related latent variables can be extracted, upon which an additional probabilistic model is constructed for further information fusion and ensemble learning. Based on detailed comparative case studies, both prediction accuracy and uncertainty have been improved by the deep probabilistic model.**

***Index Terms*—Deep learning, industrial data analytics, partial least squares (PLSs), predictive modeling, probabilistic latent variable model, residual modeling.**

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Zhiqiang Ge and Duxin Chen are with the School of Mathematics, Southeast University, Nanjing 210096, China (e-mail: zhiqiang.ge@hotmail.com; chendx@seu.edu.cn).

Wenwu Yu is with the School of Mathematics, Southeast University, Nanjing 210096, China, and also with the Purple Mountain Laboratories, Nanjing 211111, China (e-mail: wwyu@seu.edu.cn).

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I. Introduction

In the past ten years, data analytics have played an important role in Industry 4.0 by providing valuable information mined from industrial big data [1], [2], [3]. Typical application scenarios for industrial data analytics include process monitoring, soft sensing of key parameters, fault diagnosis, machine health management, remaining useful life prediction, key performance indicator prediction, and optimization [4], [5], [6], [7], [8]. To support these application scenarios, various data-driven modeling approaches need to be constructed of which predictive modeling is one of the most popular methods. For example, to provide online estimation of some difficult-to-measure variables for quality control, an effective predictive model is required to make real-time predictions from other easy-to-measure process variables. A similar predictive model is also desired for remaining useful life prediction of a key equipment in the industrial processes, to ensure the implementation of prognostic health management [9], [10], [11], [12].

For the purpose of industrial data predictive modeling, many statistical and machine learning methods have been introduced and used in the past decades. As well, deep learning has recently become popular [13], [14], [15], [16], [17], [18]. According to the search results from both Google Scholar and Web of Science, although more complicated models (shallow and deep) have been introduced, the linear regression models still play a dominant role in predictive modeling of practical industrial data analytics, such as least square regression, principal component regression, and partial least squares (PLSs) regression. Due to their simple and transparent model structures, they are more likely to provide reliable, explainable, and trustworthy results. These are the main reasons why the simple linear models are still very popular in the current ages of deep learning and artificial intelligence.

With more sensors installed in modern industrial processes, the number of measured variables has grown rapidly, which makes data mining and analytics more complicated. To alleviate this problem, latent variable models use a good balance between dimensionality reduction and modeling performance for predictive data analytics [19]. Principal component analysis (PCA) and PLSs are two common methods that have been widely studied in the past years. By projecting the original high-dimensional process variables onto a low-dimensional feature space, both analytics and regression modeling become much easier and more efficient. By considering the stochastic nature of those measured variables in practical industrial processes,

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another type of latent variable model may be fit for predictive analytics, which is known as a probabilistic latent variable model. Some representatives include probabilistic PCA, probabilistic PCR, and probabilistic PLS [20], [21], [22], [23], [24], [25]. Compared to the basic PCA and PLS models that are typically developed through data projection and coordinate transformation, probabilistic latent variable models are mostly built upon generative model structures. By assuming prior probabilities for latent variables, continuous inferences of their posterior probabilities can be implemented given the observed data based on which, the optimized probabilistic model structure can be determined. As a result, the probabilistic latent variable model can naturally deal with missing data in the model training process. Model selection can also be incorporated into the learning algorithm. As well, the generative form of the probabilistic latent variable model is more flexible and straightforward for its extension to more complex models, to obtain higher capacities and prediction performances.

The idea of deep learning has been considered as the most effective way to increase the complexity of a data-driven model. At the same time, however, several drawbacks also arise such as high computational burden, cumbersome parameter tuning, and weak model interpretability, which creates obstacles in the practical industrial application of these deep learning methods. Recently, several lightweight forms of deep learning models have been developed for industrial data analytics, i.e., deep PCA and deep PLS, which are actually combinations of deep learning and latent variable models [26], [27], [28], [29], [30], [31], [32]. Due to the lightweight model structure, fewer tuning parameters, and a relatively smaller training dataset, those deep latent-variable models seem quite suitable and useful in various industrial application scenarios for data analytics.

Inspired by the idea of lightweight deep learning implemented using the PLS model [32], this article extends the probabilistic PLS model to its deep learning form. A deep residual learning algorithm is proposed based on the generative structure of the probabilistic PLS model, with a well-designed expectation-maximization learning strategy for parameter optimization. Furthermore, a soft sensor is constructed based on the newly developed deep probabilistic PLS model for online prediction of key performance indicators in industrial processes.

II. A Brief Introduction to the Probabilistic PLS Model

The basic idea of a probabilistic PLS model is to use one part of the extracted latent variables to build the relationship between input and output variables, while another part of the latent variable is only used to explain the input variables. Based on the generative modeling framework, the probabilistic PLS model is as follows [20]:

$$\begin{aligned}\mathbf{x} &= \boldsymbol{\mu}_x + \mathbf{P}\mathbf{t}^s + \mathbf{Q}\mathbf{t}^u + \mathbf{e}_x \\ \mathbf{y} &= \boldsymbol{\mu}_y + \mathbf{C}\mathbf{t}^s + \mathbf{e}_y\end{aligned} \tag{1}$$

where $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$ are the mean values of the input variables $\mathbf{x}$ and the output variables $\mathbf{y}$, $\mathbf{P} \in R^{m\times q_s}$, $\mathbf{C} \in R^{r\times q_s}$ and $\mathbf{Q} \in R^{m\times q_b}$ are loading matrices, $\mathbf{t}^s \in R^{q_s\times 1}$ is the supervised latent variable vector that is used to connect the information of $\mathbf{x}$ and $\mathbf{y}$, $\mathbf{t}^u \in R^{q_u\times 1}$ is the unsupervised latent variable vector that is only used to explain the information of $\mathbf{x}$, $\mathbf{e}_x \in R^{m\times 1}$ and $\mathbf{e}_y \in R^{r\times 1}$ are the measurement noises of $\mathbf{x}$ and $\mathbf{y}$.

In this model, it is assumed that both latent variables and measurement noises are normally distributed. Thus, the prior probabilities for supervised and unsupervised latent variables are given as $p(\mathbf{t}^s) = N(0, \mathbf{I})$, $p(\mathbf{t}^u) = N(0, \mathbf{I})$, and the probabilities for the two measurement noises are $p(\mathbf{e}_x) = N(0, \boldsymbol{\Sigma}_x)$, and $p(\mathbf{e}_y) = N(0, \boldsymbol{\Sigma}_y)$, where $\boldsymbol{\Sigma}_x = \text{diag}\{\sigma^2_{x,i}\}_{i=1,2,\cdots,m}$, $\boldsymbol{\Sigma}_y = \text{diag}\{\sigma^2_{y,j}\}_{j=1,2,\cdots,r}$. It can be noted that different noise variances have been assumed for the different process variables.

III. Deep Residual Probabilistic PLS Modeling Framework

In this section, a deep residual learning algorithm is proposed for the probabilistic PLS model, combined with a detailed description of an efficient EM algorithm for parameter optimization. After that, an industrial soft sensor will be designed for online prediction of key variables to support the development of a quality control system. Suppose we are given the data information of the input and output process variables $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n]^T \in R^{n\times m}$ and $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_n]^T \in R^{n\times r}$. The first step to develop the deep probabilistic PLS model is to construct an initial probabilistic PLS model from this original dataset, given as

$$\begin{aligned}\mathbf{x} &= \mathbf{P}^{(1)}\mathbf{t}^{s^{(1)}} + \mathbf{Q}^{(1)}\mathbf{t}^{u^{(1)}} + \mathbf{e}_x^{(1)} \\ \mathbf{y} &= \mathbf{C}^{(1)}\mathbf{t}^{s^{(1)}} + \mathbf{e}_y^{(1)}\end{aligned} \tag{2}$$

where symbol (1) indicates the first layer of the deep learning model. For simplicity, the mean values of both input and the output variables have been ignored in the above model structure.

Based on the Gaussian distributions of latent variables and measurement noises, the conditional probability distributions of both input and output variables $\mathbf{x}$ and $\mathbf{y}$ can be formed as $p(\mathbf{x}|\mathbf{t}^{s^{(1)}}, \mathbf{t}^{u^{(1)}}, \mathbf{P}^{(1)}, \mathbf{Q}^{(1)}, \boldsymbol{\Sigma}_x^{(1)}) = N(\mathbf{P}^{(1)}\mathbf{t}^{s^{(1)}} + \mathbf{Q}^{(1)}\mathbf{t}^{u^{(1)}}, \boldsymbol{\Sigma}_x^{(1)})$ and $p(\mathbf{y}|\mathbf{t}^{s^{(1)}}, \mathbf{C}^{(1)}, \boldsymbol{\Sigma}_y^{(1)}) = N(\mathbf{C}^{(1)}\mathbf{t}^{s^{(1)}}, \boldsymbol{\Sigma}_y^{(1)})$, respectively. By integrating out the latent variables, the marginal distribution of $\mathbf{x}$ and $\mathbf{y}$ can be determined as

$$\begin{aligned}&p\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_x^{(1)}\\ \boldsymbol{\Sigma}_y^{(1)}\end{bmatrix}\right) \\ &= \int p\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}^{s^{(1)}}\\ \mathbf{t}^{u(1)}\end{bmatrix}, \begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}, \begin{bmatrix}\boldsymbol{\Sigma}_x^{(1)}\\ \boldsymbol{\Sigma}_y^{(1)}\end{bmatrix}\right) \\ &\quad p\left(\begin{bmatrix}\mathbf{t}^{s^{(1)}}\\ \mathbf{t}^{u(1)}\end{bmatrix}\right) d\begin{bmatrix}\mathbf{t}^{s^{(1)}}\\ \mathbf{t}^{u(1)}\end{bmatrix}.\end{aligned} \tag{3}$$

To determine the optimal value of the parameter set $\boldsymbol{\Theta}^{(1)} = \{\mathbf{P}^{(1)}, \mathbf{Q}^{(1)}, \mathbf{C}^{(1)}, \boldsymbol{\Sigma}_x^{(1)}, \boldsymbol{\Sigma}_y^{(1)}\}$ for this initial probabilistic PLS model, an efficient EM algorithm can be designed. Instead of directly maximizing the log-likelihood function of the marginal distribution for each data sample, the EM algorithm tries to

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maximize the expected complete-data log-likelihood function. Here, the complete data includes the observed input and output variables $\mathbf{x}$, $\mathbf{y}$ and the latent variables $\mathbf{t}^s$, $\mathbf{t}^u$, where the latent variables are considered as missing data in the design of the EM algorithm. Therefore, with respect to the latent variables, the expected value of the complete-data log-likelihood function is given in (4), shown at the bottom of the page, where $\mathbf{\Theta}^{(1)}_{\text{old}}$ is the parameter set calculated in the last step. The EM algorithm is an iterative learning process, which includes two steps: the E-step and the M-step. In the E-step, the objective is to calculate the expected value of the complete-data log-likelihood function, which is determined by the posterior probability of the latent variables given the observed input and output variables. According to Bayes' rule, the posterior probability of the latent variables can be calculated as

$$p\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)\propto p\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)\times p\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\right). \tag{5}$$

Since both the supervised and unsupervised latent variables are Gaussian, their posterior distributions are also Gaussian. As a result, the information of the posterior distribution can be determined by two sufficient statistics (6) and (7), shown at the bottom of the page.

In the M-step of the EM algorithm, the objective is to update the model parameters by maximizing the expected complete-data log likelihood. Thus, the new parameters can be calculated by setting the partial derivative of the expected complete-data log likelihood function with respect to each parameter as zero. The detailed results are given in (8)â(12), shown at the bottom of the page.

By iteratively updating those two main steps, the optimal value of the parameter set for the first layer of the deep model $\mathbf{\Theta}^{(1)}=\{\mathbf{P}^{(1)},\mathbf{Q}^{(1)},\mathbf{C}^{(1)},\mathbf{\Sigma}^{(1)}_x,\mathbf{\Sigma}^{(1)}_y\}$ can be determined as soon as the EM algorithm converges.

$$\begin{aligned}E\left[L\left(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)}\right)\right]&=\sum_{i=1}^{n}\int p\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)\ln\left[p\left(\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix},\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\mathbf{\Theta}^{(1)}\right)\right]d\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\\&=\sum_{i=1}^{n}\int p\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)\ln\left[p\left(\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix},\mathbf{\Theta}^{(1)}\right)p\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\mathbf{\Theta}\right)\right]d\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\end{aligned} \tag{4}$$

$$E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)=\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}^T\left(\begin{bmatrix}\mathbf{\Sigma}^{(1)}_x&0\\ 0&\mathbf{\Sigma}^{(1)}_y\end{bmatrix}+\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}&\mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}^T\right)^{-1}\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix} \tag{6}$$

$$\begin{aligned}E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\mathbf{t}^{s(1)T}\\ \mathbf{t}^{u(1)}\mathbf{t}^{u(1)T}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)&=\mathbf{I}-\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}^T\left(\begin{bmatrix}\mathbf{\Sigma}^{(1)}_x&0\\ 0&\mathbf{\Sigma}^{(1)}_y\end{bmatrix}+\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}^T\right)^{-1}\begin{bmatrix}\mathbf{P}^{(1)}&\mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)}&\mathbf{0}\end{bmatrix}\\&\quad+E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right)E^T\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\mathbf{\Theta}^{(1)}_{\text{old}}\right).\end{aligned} \tag{7}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)})]}{\partial\mathbf{P}^{(1)}}=0\Rightarrow\mathbf{P}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{x}_iE^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(1)}\mathbf{t}_i^{s(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{8}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)})]}{\partial\mathbf{Q}^{(1)}}=0\Rightarrow\mathbf{Q}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{x}_iE^T\left(\mathbf{t}_i^{u(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{u(1)}\mathbf{t}_i^{u(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{9}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)})]}{\partial\mathbf{C}^{(1)}}=0\Rightarrow\mathbf{C}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{y}_iE^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(1)}\mathbf{t}_i^{s(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{10}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)})]}{\partial\mathbf{\Sigma}^{(1)}_x}=0\Rightarrow\mathbf{\Sigma}^{(1)\text{new}}_x=\frac{1}{n}\text{diag}\left\{\sum_{i=1}^{n}\left[\mathbf{x}_i\mathbf{x}_i^T-\begin{bmatrix}\mathbf{P}^{(1)\text{new}}&\mathbf{Q}^{(1)\text{new}}\end{bmatrix}E\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{x}_i^T\right]\right\} \tag{11}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\mathbf{\Theta}^{(1)})]}{\partial\mathbf{\Sigma}^{(1)}_y}=0\Rightarrow\mathbf{\Sigma}^{(1)\text{new}}_y=\frac{1}{n}\text{diag}\left\{\sum_{i=1}^{n}\left[\mathbf{y}_i\mathbf{y}_i^T-\mathbf{C}^{(1)\text{new}}E\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{y}_i^T\right]\right\}. \tag{12}$$

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In the second layer of the deep probabilistic PLS model, we try to capture more hidden latent variable information to explain the target output variables $\mathbf{y}$ from the residual data of the input variables $\mathbf{x}$. Therefore, the second layerwise probabilistic PLS model is constructed upon the residual data information from the first layer, that is,

$$\left[\mathbf{x}^{r(1)} = \mathbf{x} - \mathbf{P}^{(1)}\mathbf{t}^{s^{(1)}}\right] \Rightarrow \begin{bmatrix} \mathbf{x}^{r(1)} = \mathbf{P}^{(2)}\mathbf{t}^{s(2)} + \mathbf{Q}^{(2)}\mathbf{t}^{u(2)} + \mathbf{e}_x^{(2)} \\ \mathbf{y} = \mathbf{C}^{(2)}\mathbf{t}^{s(2)} + \mathbf{e}_y^{(2)} \end{bmatrix}. \quad (13)$$

For training this layerwise model, a similar EM algorithm can be designed to determine the optimal value of the parameter set in this layer. Without loss of generality, it is supposed that a total of L hidden layers have been used to construct the deep probabilistic PLS model. Thus, the expected complete-data log likelihood function is given in (14), shown at the bottom of the page, where $l = 1, 2, \cdots, L, \mathbf{X}^{r(0)} = \mathbf{X}$.

After all of the L layerwise probabilistic PLS models have been constructed, various supervised latent variables are extracted from the different hidden layers, which denote different connections between the input variables $\mathbf{x}$ and the target output variables $\mathbf{y}$. To make a further extraction for key supervised latent variables and to eliminate redundant information, an additional probabilistic PLS model is constructed for those supervised latent variables extracted from the different hidden layers. Thus, a deep latent variable vector is defined as

$$\mathbf{t}^d = [\mathbf{t}^{s(1)}\, \mathbf{t}^{s(2)}\, \cdots\, \mathbf{t}^{s(L)}]^T. \quad (15)$$

To build the relationship between the deep latent variables and the target output variables $\mathbf{y}$, the additional probabilistic PLS model is formulated as

$$\begin{aligned} \mathbf{t}^d &= \mathbf{P}^{(d)}\mathbf{t}^{s(d)} + \mathbf{Q}^{(d)}\mathbf{t}^{u(d)} + \mathbf{e}_x^{(d)} \\ \mathbf{y} &= \mathbf{C}^{(d)}\mathbf{t}^{s(d)} + \mathbf{e}_y^{(d)}. \end{aligned} \quad (16)$$

By maximizing the expected complete-data log likelihood function given in (17), the optimal parameter set of this additional probabilistic PLS model can be determined in (17), shown at the bottom of the page.

Similarly, in the E-step of the EM algorithm for this probabilistic PLS model, the posterior distribution of both supervised and unsupervised latent variables $\mathbf{t}^{s(d)}$ and $\mathbf{t}^{u(d)}$ are determined using Bayes' rule, which gives the following:

$$p\left(\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) \propto p\left(\begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix} \middle| \left[\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix}\right], \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) p\left(\left[\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix}\right]\right). \quad (18)$$

Based on the posterior probabilities, the two sufficient statistics are as follows:

In the M-step, the parameters are updated using (21)â(25), shown at the bottom of the next page.

By iteratively calculating both the E-step and the M-step, the optimal value of the parameter set for the final probabilistic PLS model can be determined.

For the development of an industrial soft sensor for the proposed model, some easy-to-measure process variables are used as the input of the soft sensor. The difficult-to-measure

$$\begin{aligned} E[L(\mathbf{X}^{r(l-1)}, \mathbf{Y}, \boldsymbol{\Theta}^{(l)})] &= \sum_{i=1}^{n} \int p\left(\begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{x}_i^{r(l-1)} \\ \mathbf{y}_i\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(l)}\right) \ln\left[p\left(\begin{bmatrix}\mathbf{x}_i^{r(l-1)} \\ \mathbf{y}_i\end{bmatrix}, \begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \middle| \boldsymbol{\Theta}^{(l)}\right)\right] d\begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \\ &= \sum_{i=1}^{n} \int p\left(\begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{x}_i^{r(l-1)} \\ \mathbf{y}_i\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(l)}\right) \ln\left[p\left(\begin{bmatrix}\mathbf{x}_i^{r(l-1)} \\ \mathbf{y}_i\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix}, \boldsymbol{\Theta}^{(l)}\right) p\left(\begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \middle| \boldsymbol{\Theta}\right)\right] d\begin{bmatrix}\mathbf{t}_i^{s(l)} \\ \mathbf{t}_i^{u(l)}\end{bmatrix} \end{aligned} \quad (14)$$

$$\begin{aligned} E\left[L\left(\mathbf{T}^d, \mathbf{Y}, \boldsymbol{\Theta}^{(d)}\right)\right] &= \sum_{i=1}^{n} \int p\left(\begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}_i^{d} \\ \mathbf{y}_i\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) \ln\left[p\left(\begin{bmatrix}\mathbf{x}_i \\ \mathbf{y}_i\end{bmatrix}, \begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix} \middle| \boldsymbol{\Theta}^{(l)}\right)\right] d\begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix} \\ &= \sum_{i=1}^{n} \int p\left(\begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}_i^{d} \\ \mathbf{y}_i\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) \ln\left[p\left(\begin{bmatrix}\mathbf{t}_i^{d} \\ \mathbf{y}_i\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix}, \boldsymbol{\Theta}^{(l)}\right) p\left(\begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix} \middle| \boldsymbol{\Theta}\right)\right] d\begin{bmatrix}\mathbf{t}_i^{s(d)} \\ \mathbf{t}_i^{u(d)}\end{bmatrix}. \end{aligned} \quad (17)$$

$$E\left(\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) = \begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix}^T \left(\begin{bmatrix}\boldsymbol{\Sigma}_x^{(d)} & 0 \\ 0 & \boldsymbol{\Sigma}_y^{(d)}\end{bmatrix} + \begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix}^T\right)^{-1} \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix} \quad (19)$$

$$\begin{aligned} E\left(\begin{bmatrix}\mathbf{t}^{s(d)} & \mathbf{t}^{s(d)T} \\ \mathbf{t}^{u(d)} & \mathbf{t}^{u(d)T}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(1)}\right) &= \mathbf{I} - \begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix}^T \left(\begin{bmatrix}\boldsymbol{\Sigma}_x^{(d)} & 0 \\ 0 & \boldsymbol{\Sigma}_y^{(d)}\end{bmatrix} + \begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(1)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix}^T\right)^{-1} \begin{bmatrix}\mathbf{P}^{(d)} & \mathbf{Q}^{(d)} \\ \mathbf{C}^{(d)} & \mathbf{0}\end{bmatrix} \\ &\quad + E\left(\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right) E^T\left(\begin{bmatrix}\mathbf{t}^{s(d)} \\ \mathbf{t}^{u(d)}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{t}^d \\ \mathbf{y}\end{bmatrix}, \boldsymbol{\Theta}_{\text{old}}^{(d)}\right). \end{aligned} \quad (20)$$

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key performance indicators or quality-related variables are the output of the soft sensor. By constructing the deep residual probabilistic PLS model between these two different types of variables, online predictions can be made by the soft sensor.

For a new data sample $\mathbf{x}_{\text{new}}$, the posterior probabilities of both supervised and unsupervised latent variables in different layers of the deep model can be calculated in the first step

$$
\begin{aligned}
&p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(1)}\\ \mathbf{t}_{\text{new}}^{u(1)}\end{bmatrix}\middle|\mathbf{x}_{\text{new}},\boldsymbol{\Theta}^{(2)}\right)\propto p\left(\mathbf{x}_{\text{new}}\middle|\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(1)}\\ \mathbf{t}_{\text{new}}^{u(1)}\end{bmatrix},\boldsymbol{\Theta}^{(1)}\right)\\
&p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(1)}\\ \mathbf{t}_{\text{new}}^{u(1)}\end{bmatrix}\middle|\boldsymbol{\Theta}^{(1)}\right)p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(2)}\\ \mathbf{t}_{\text{new}}^{u(2)}\end{bmatrix}\middle|\mathbf{x}_{\text{new}}^{r(1)},\boldsymbol{\Theta}^{(2)}\right)\\
&\propto p\left(\mathbf{x}_{\text{new}}^{r(1)}\middle|\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(1)}\\ \mathbf{t}_{\text{new}}^{u(1)}\end{bmatrix},\boldsymbol{\Theta}^{(2)}\right)p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(1)}\\ \mathbf{t}_{\text{new}}^{u(1)}\end{bmatrix}\middle|\boldsymbol{\Theta}^{(2)}\right)\\
&\qquad\vdots\\
&p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(L)}\\ \mathbf{t}_{\text{new}}^{u(L)}\end{bmatrix}\middle|\mathbf{x}_{\text{new}}^{r(L-1)},\boldsymbol{\Theta}^{(L)}\right)\propto p\left(\mathbf{x}_{\text{new}}^{r(L-1)}\middle|\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(L)}\\ \mathbf{t}_{\text{new}}^{u(L)}\end{bmatrix},\boldsymbol{\Theta}^{(L)}\right)\\
&p\left(\begin{bmatrix}\mathbf{t}_{\text{new}}^{s(L)}\\ \mathbf{t}_{\text{new}}^{u(L)}\end{bmatrix}\middle|\boldsymbol{\Theta}^{(L)}\right).
\end{aligned}
\tag{26}
$$

To provide a point prediction for the target variables, the mean values of the latent variables in each hidden layer can be determined using

$$
\begin{aligned}
&\begin{bmatrix}\bar{\mathbf{t}}_{\text{new}}^{s(1)}\\ \bar{\mathbf{t}}_{\text{new}}^{u(1)}\end{bmatrix}=[\mathbf{P}^{(1)}\ \mathbf{Q}^{(1)}]^T(\boldsymbol{\Sigma}_x^{(1)}+[\mathbf{P}^{(1)}\ \mathbf{Q}^{(1)}][\mathbf{P}^{(1)}\ \mathbf{Q}^{(1)}]^T)^{-1}\mathbf{x}_{\text{new}}\\
&\begin{bmatrix}\bar{\mathbf{t}}_{\text{new}}^{s(2)}\\ \bar{\mathbf{t}}_{\text{new}}^{u(2)}\end{bmatrix}=[\mathbf{P}^{(2)}\ \mathbf{Q}^{(2)}]^T(\boldsymbol{\Sigma}_x^{(2)}+[\mathbf{P}^{(2)}\ \mathbf{Q}^{(2)}][\mathbf{P}^{(2)}\ \mathbf{Q}^{(2)}]^T)^{-1}\mathbf{x}_{\text{new}}^{r(1)}\\
&\qquad\vdots\\
&\begin{bmatrix}\bar{\mathbf{t}}_{\text{new}}^{s(L)}\\ \bar{\mathbf{t}}_{\text{new}}^{u(L)}\end{bmatrix}=[\mathbf{P}^{(L)}\ \mathbf{Q}^{(L)}]^T(\boldsymbol{\Sigma}_x^{(L)}+[\mathbf{P}^{(L)}\ \mathbf{Q}^{(L)}]\\
&\quad\times[\mathbf{P}^{(L)}\ \mathbf{Q}^{(L)}]^T)^{-1}\mathbf{x}_{\text{new}}^{r(L-1)}.
\end{aligned}
\tag{27}
$$

Then, based on the extracted supervised latent variables which are used to explain the target variables, the deep latent variables for the new data sample $\mathbf{x}_{\text{new}}$ are formulated as

$$
\mathbf{t}_{\text{new}}^{d}=\begin{bmatrix}\bar{\mathbf{t}}_{\text{new}}^{s(1)}\ \bar{\mathbf{t}}_{\text{new}}^{s(2)}\ \cdots\ \bar{\mathbf{t}}_{\text{new}}^{s(L)}\end{bmatrix}^T. \tag{28}
$$

Finally, the value of point prediction for the new data sample $\mathbf{x}_{\text{new}}$ can be determined as

$$
\begin{aligned}
&\begin{bmatrix}\bar{\mathbf{t}}_{\text{new}}^{s(d)}\\ \bar{\mathbf{t}}_{\text{new}}^{u(d)}\end{bmatrix}=[\mathbf{P}^{(d)}\ \mathbf{Q}^{(d)}]^T(\boldsymbol{\Sigma}_x^{(d)}+[\mathbf{P}^{(d)}\ \mathbf{Q}^{(d)}]\\
&\quad\times[\mathbf{P}^{(d)}\ \mathbf{Q}^{(d)}]^T)^{-1}\mathbf{t}_{\text{new}}^{d}\\
&\qquad\qquad\Downarrow\\
&\hat{\mathbf{y}}_{\text{new}}=\mathbf{C}^{(d)}\bar{\mathbf{t}}_{\text{new}}^{s(d)}.
\end{aligned}
\tag{29}
$$

The complete procedure for the deep residual probabilistic PLS model is shown in Fig. 1.

IV. Industrial Case Studies

A. Debutanizer Column

The debutanizer column is a widely used industrial system for performance evaluation of various data-driven soft sensors [33], [34], [35], [36], [37], [38], [39]. This column is an important part of the desulfuring and naphtha splitter plant. To control the performance of this column, real-time acquisition of the butane content is very important, which implies that this variable can be considered to be a key performance indicator or quality variable. In practice, however, the butane content is difficult to measure online, compared to other routinely recorded process variables such as the temperatures, pressures, and steam flows. Therefore, a soft sensor is required for online estimation of this key variable. A detailed flowchart of

$$
\frac{\partial E[L(\mathbf{T}^d,\mathbf{Y},\boldsymbol{\Theta}^{(d)})]}{\partial\mathbf{P}^{(d)}}=0\Rightarrow\mathbf{P}^{(\text{d})\text{new}}=\left[\sum_{i=1}^{n}\mathbf{t}_i^dE^T\left(\mathbf{t}_i^{s(d)}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(d)}\mathbf{t}_i^{s(d)^T}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{21}
$$

$$
\frac{\partial E[L(\mathbf{T}^d,\mathbf{Y},\boldsymbol{\Theta}^{(d)})]}{\partial\mathbf{Q}^{(d)}}=0\Rightarrow\mathbf{Q}^{(d)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{t}_i^dE^T\left(\mathbf{t}_i^{u(d)}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{u(d)}\mathbf{t}_i^{u(d)^T}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{22}
$$

$$
\frac{\partial E[L(\mathbf{T}^d,\mathbf{Y},\boldsymbol{\Theta}^{(d)})]}{\partial\mathbf{C}^{(d)}}=0\Rightarrow\mathbf{C}^{(d)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{y}_iE^T\left(\mathbf{t}_i^{s(d)}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(d)}\mathbf{t}_i^{s(d)^T}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{23}
$$

$$
\frac{\partial E[L(\mathbf{T}^d,\mathbf{Y},\boldsymbol{\Theta}^{(d)})]}{\partial\boldsymbol{\Sigma}_x^{(d)}}=0\Rightarrow\boldsymbol{\Sigma}_x^{(d)\text{new}}=\frac{1}{n}\text{diag}\left\{\sum_{i=1}^{n}\left[\mathbf{t}_i^d\mathbf{t}_i^{dT}-\begin{bmatrix}\mathbf{P}^{(d)\text{new}} & \mathbf{Q}^{(d)\text{new}}\end{bmatrix}E\left(\begin{bmatrix}\mathbf{t}_i^{s(d)}\\ \mathbf{t}_i^{u(d)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{t}_i^{dT}\right]\right\} \tag{24}
$$

$$
\frac{\partial E[L(\mathbf{T}^d,\mathbf{Y},\boldsymbol{\Theta}^{(d)})]}{\partial\boldsymbol{\Sigma}_y^{(d)}}=0\Rightarrow\boldsymbol{\Sigma}_y^{(d)\text{new}}=\frac{1}{n}\text{diag}\left\{\sum_{i=1}^{n}\left[\mathbf{y}_i\mathbf{y}_i^T-\mathbf{C}^{(d)\text{new}}E\left(\mathbf{t}_i^{s(d)}\middle|\begin{bmatrix}\mathbf{t}_i^d\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{y}_i^T\right]\right\}. \tag{25}
$$

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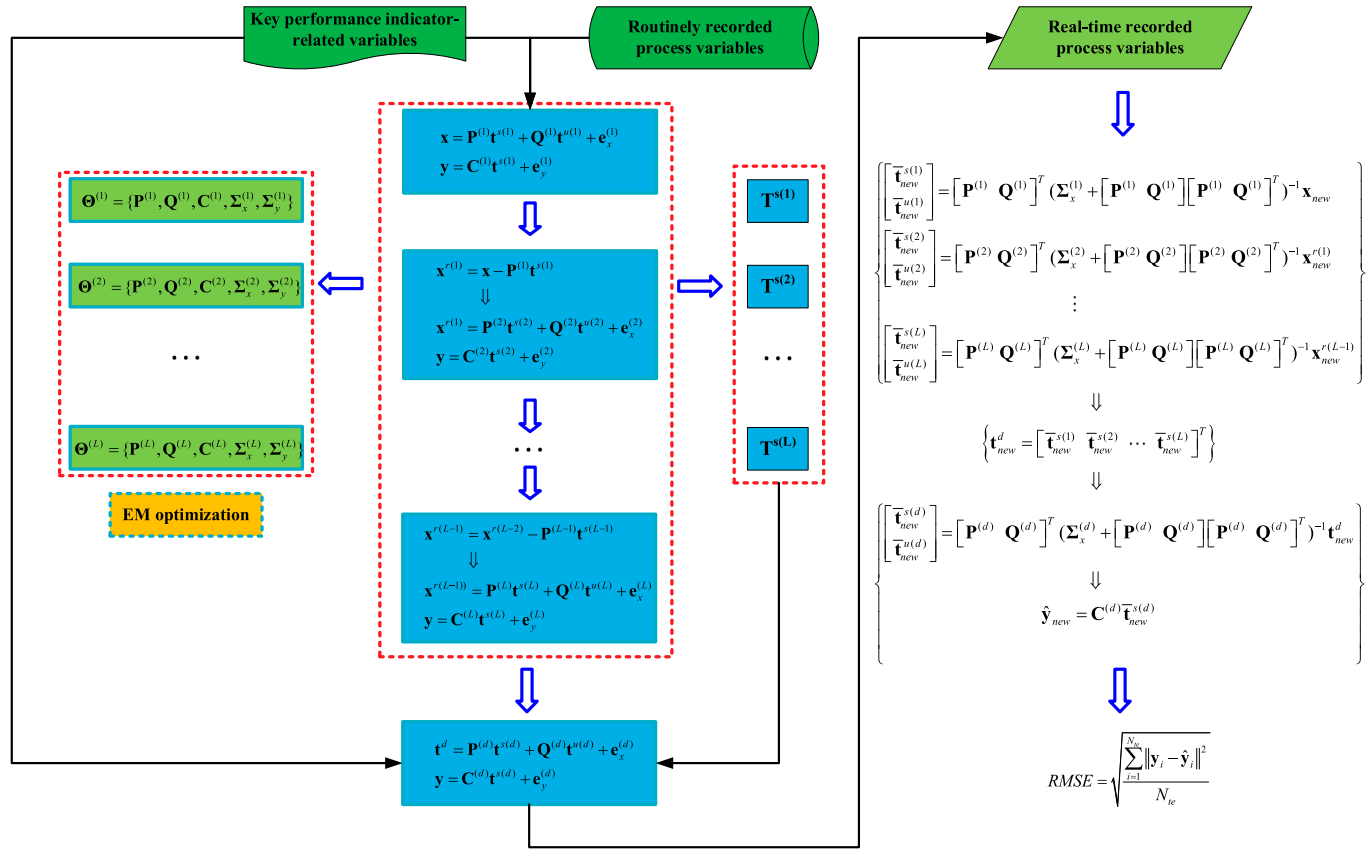


Fig. 1. Model training and application process of the deep residual probabilistic PLS model.

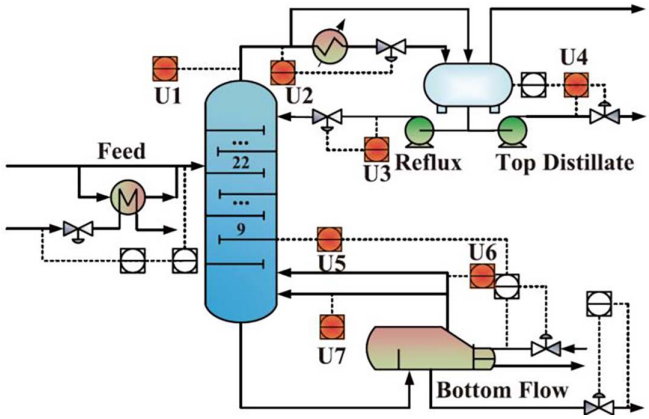


Fig. 2. Flowchart of the debutanizer column [33].

TABLE I
INPUT VARIABLES IN THE DEBUTANIZER COLUMN

Input Variables	Descriptions
x_1	Top temperature
x_2	Top pressure
x_3	Reflux flow
x_4	Flow to next process
x_5	Temperature on the 6th tray
x_6	Bottom temperature #1
x_7	Bottom temperature #2

the debutanizer column is shown in Fig. 2 [33]. For online prediction of the butane content in the process, seven related process variables are selected as the inputs for the soft sensor. The detailed descriptions of the variables are provided in Table I.

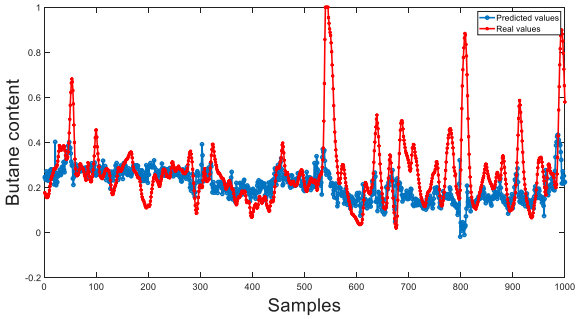
For the development of the deep residual probabilistic PLS model, a training dataset has been collected in advance, which contains 1000 data samples. Another dataset which also contains 1000 data samples is used for performance testing of the soft sensor. For comparison, the basic probabilistic PLS model is also used for soft sensing of the butane content in this process. To evaluate the prediction performance of these two different soft sensors, the root mean squared error (RMSE) index is used, which is defined as

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N_{te}} \|\mathbf{y}_i - \hat{\mathbf{y}}_i\|^2}{N_{te}}} \tag{30}$$

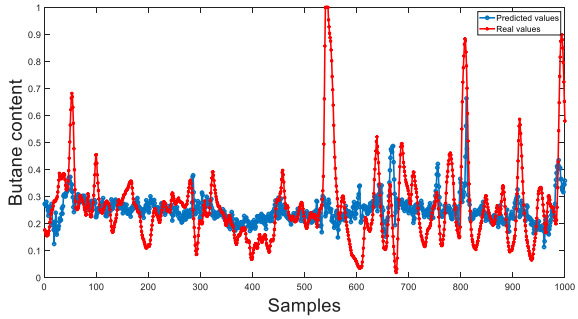
where $i = 1, 2, \cdots, N_{te}$, N_{te} is the total number of testing data samples.

In this case study, the number of hidden layers in the deep residual probabilistic PLS model is selected as 3, while the numbers of supervised and unsupervised latent variables are selected as 3 and 2 in both the basic and layerwise probabilistic PLS models. In the final probabilistic PLS model, the numbers of supervised and unsupervised latent variables are selected as

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(a)



(b)

Fig. 3. Prediction results for the testing dataset: (a) PPLS; and (b) deep PPLS.

2 and 2, respectively, to further compress the data information in both subspaces of supervised and unsupervised latent variables. For the probabilistic PLS model, it is quite difficult to evaluate the importance or value of each extracted latent variable. In other words, one particular latent variable may have quite different contributions in different modeling tasks. Even for the same predictive modeling task, different latent variables have shown diverse correlations to different target variables. To provide a relatively stable model structure, the numbers of both supervised and unsupervised latent variables have been determined through the cross-validation approach. The determination of the number of hidden layers in the deep model is a well-known open question in the research field. Although some possible data-driven optimization algorithms can be developed to automate this process, their effectiveness varies from case to case. Therefore, in this article, the number of hidden layers used for the construction of the deep probabilistic PLS model is also determined using the cross-validation method.

Detailed prediction results for the 1000 testing data samples by the two probabilistic-model-based soft sensors are shown in Fig. 3. One can see that the prediction accuracy has been significantly improved by the deep model. According to the value of the RMSE, the deep learning model has reduced the RMSE by 20% compared with the baseline model. Furthermore, the value of the R^2 index has increased by more than 30%, which means the predicted values of the testing data samples are more correlated with the real values. To further show how the two models compared with each other, the parity plot for both models is shown in Fig. 4. As can be seen from this figure, the values predicted by the deep PPLS model are more correlated

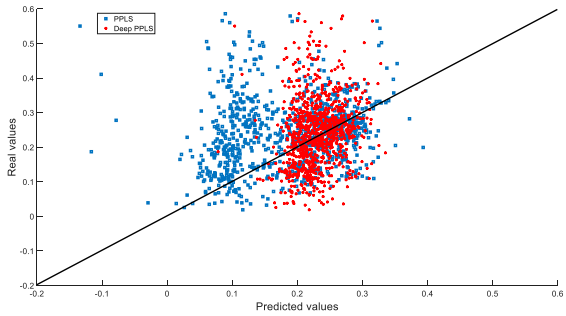


Fig. 4. Parity plot showing the results of the two models.

TABLE II
COMPARISONS OF RMSE VALUES FOR DIFFERENT MODELS

Models	Layer #1	Layer #2	Layer #3	Deep PPLS
RMSE	0.1763	0.1560	0.1710	0.1529

with the real values of the testing data samples than those provided by the basic PPLS model, which is consistent with the performance improvement in the R^2 index. For comparison with the basic PLS model, the performances obtained from the two shallow models (deterministic and probabilistic) are similar. As shown in previous studies, the probabilistic PLS model should be more favorable in those application scenarios with significant data uncertainties or measurement noises.

To examine the quality of the supervised latent variables in each hidden layer, the performance of different layerwise probabilistic PLS models is compared to the final probabilistic PLS model. The detailed results are shown in Table II. As can be seen from this figure, the best prediction result has been provided by the second layerwise model of all three layerwise models. It is quite difficult to evaluate the quality of those extracted latent variables in different hidden layers, because the performance of different layerwise predictive models may vary from case to case. Therefore, it is necessary that an additional predictive model could be constructed, not only for the purpose of information fusion but also to reduce the uncertainty in the prediction of new data samples.

Fig. 5 shows prediction results of the two probabilistic models with corresponding uncertainties for the first 100 testing samples. It is very clear that the deep residual probabilistic PLS model has greatly reduced the prediction variance, compared to the result shown in Fig. 6(a) by the basic probabilistic PLS model. Furthermore, the prediction uncertainty in terms of estimated variance is evaluated using different layerwise models and the final probabilistic PLS model. The results are shown in Table III. Based on those results, it can be inferred that the additional probabilistic PLS model does have a positive effect on reducing the prediction uncertainty. Finally, sensitivity analysis for the parameters in the deep probabilistic PLS model is provided. Table IV presents the change of RMSE values for different numbers of hidden layers used in the deep model structure. It can be seen that the prediction performance of the model is quite sensitive to the number of hidden layers. In

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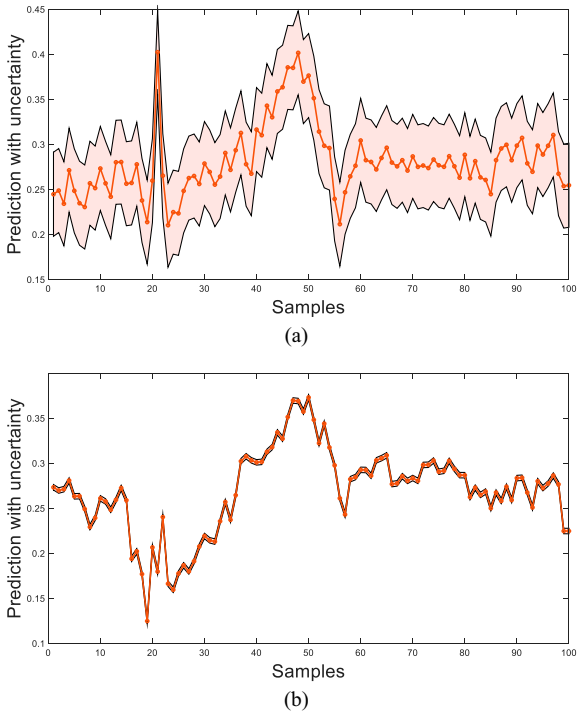


Fig. 5. Prediction results with uncertainties for the first 100 samples: (a) PPLS; and (b) deep PPLS.

TABLE III
COMPARISONS OF PREDICTION VARIANCES FOR DIFFERENT MODELS

Models	Layer #1	Layer #2	Layer #3	Deep PPLS
Prediction variance	0.0009	0.001	0.0004	$3.36*10^{-6}$

TABLE IV
RMSE VALUES FOR DIFFERENT NUMBERS OF HIDDEN LAYERS USED IN THE DEEP MODEL

Numbers of Hidden Layers	1	2	3	4	5
RMSE	0.1713	0.1895	0.1528	0.1641	0.1778
Numbers of Hidden Layers	6	7	8	9	10
RMSE	0.1949	0.1557	0.1618	0.1667	0.1561

TABLE V
RMSE VALUES FOR DIFFERENT NUMBERS OF SELECTED LATENT VARIABLES

Number of Latent Variables	3	4	5	6
PPLS	0.1587	0.1532	0.1927	0.1665
Deep PPLS	0.1742	0.1453	0.1662	0.1635

practical applications, however, the selection of this parameter needs to be considered together with the computation burden of the deep model, since the complexity of the model will gradually increase as the number of hidden layers increases. In addition, the RMSE values of the two models for different numbers of selected latent variables are shown in Table V. One can see that the best performances have been obtained for both models when the number is selected as 4.

B. Cement Clinker Production Process

Cement is an important building material. Its production process contains several operating units, including the preheater, the calciner, the rotary kiln, and the grate cooler. A flowchart of this process is shown in Fig. 6, which is from a real cement company in East China. In cement production, the content of the free calcium oxide (f-CaO) in the clinker is a critical factor, which significantly affects the quality of the final product. Therefore, a quality control system is required to monitor this variable. In practice, however, there is currently no effective instrument that can provide continuous measurements for this important variable. As a result, the value of f-CaO content in the clinker has to be measured offline through manual sampling and laboratory analysis, which are both time consuming and expensive. Therefore, in this case, a data-driven soft sensor is particularly useful, which can provide real-time estimations for the content of f-CaO in the clinker.

To predict the f-CaO content in the cement clinker, all of the 103 routinely recorded process variables are used as the input for the soft sensor. For model training and performance evaluation, the complete dataset which contains 600 effective data samples is partitioned into two equal parts. To develop the deep probabilistic PLS model, the number of hidden layers in this case study is selected as 5, while the numbers of supervised and unsupervised latent variables are selected as 10 and 5 in both the basic and layerwise probabilistic PLS models. In the final probabilistic PLS model, the numbers of supervised and unsupervised latent variables are selected as 8 and 4, respectively. Similar to the first case study, those parameters were also determined using the cross-validation method.

Detailed prediction results of the f-CaO content for the 300 testing data samples by the shallow and deep probabilistic model-based soft sensors are shown in Fig. 7. As can be seen from Fig. 7(a), the basic PPLS model has some very bad prediction points which are negative that are highlighted by two ellipses. Again, by comparing the values of the RMSE index, the soft sensing performance has been improved by 10% using the deep probabilistic model, while the value of the R^2 index has increased by more than 15%. The parity plot for both models is shown in Fig. 8.

Similarly, the soft sensing performances of different layerwise probabilistic PLS models are compared to the final probabilistic PLS model in Table VI. As can be seen, although the final probabilistic PLS model has obtained the best prediction result, different layerwise models have shown quite different soft sensing performances. Therefore, again, the quality of the supervised latent variables in each hidden layer of the deep model is difficult to evaluate and predict. First, the probabilistic model itself has some uncertainties in the process of parameter learning. Second, the number of supervised latent variables selected for each hidden layer is a trial-and-error process, which may introduce additional uncertainty for soft sensing. As well, the deep learning model structure has its own layer-by-layer uncertainty in feature extraction.

However, with the additional probabilistic PLS model developed using the supervised latent variables obtained in different

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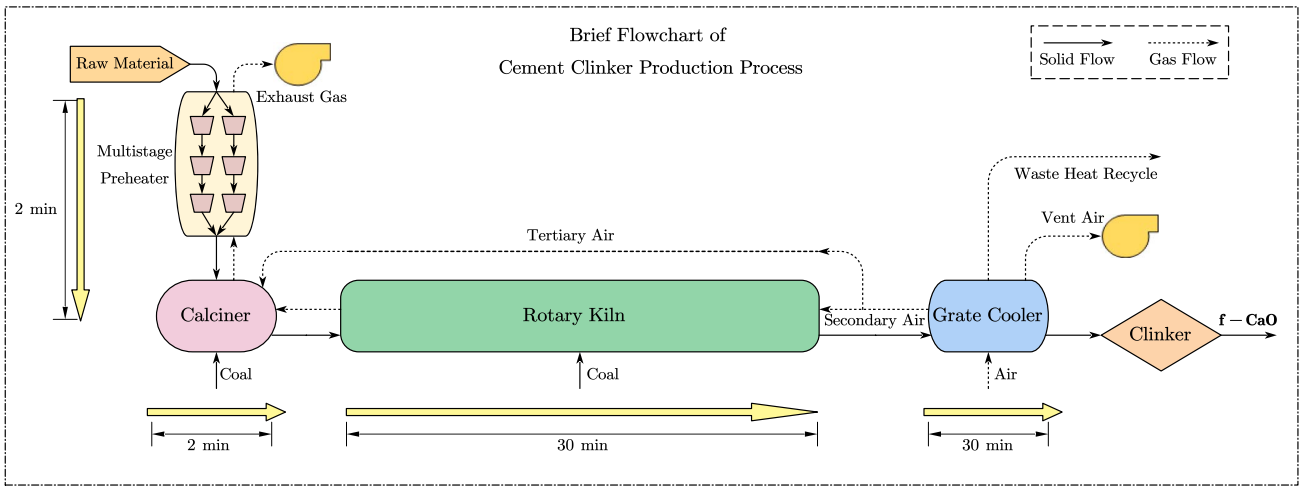


Fig. 6. Brief flowchart of the cement production.

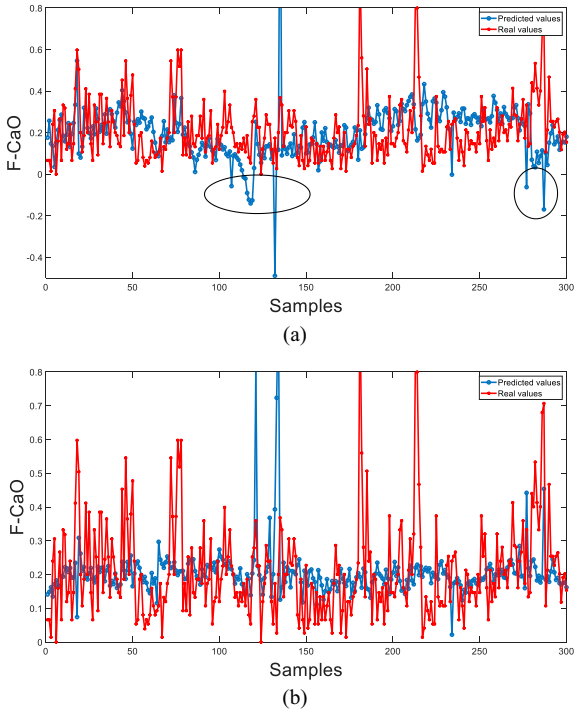


Fig. 7. Prediction results for the testing dataset: (a) PPLS; and (b) deep PPLS.

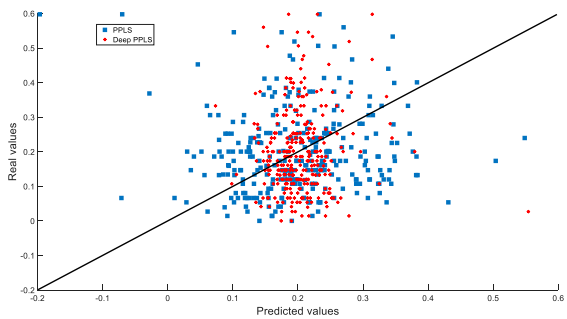


Fig. 8. Parity plot for the two models for the f-CaO.

TABLE VI
COMPARISONS OF RMSE VALUES FOR DIFFERENT MODELS

Models	Layer #1	Layer #2	Layer #3	Layer #4	Layer #5	Deep PPLS
RMSE	0.1569	0.1458	0.4749	0.6068	0.1509	0.1416

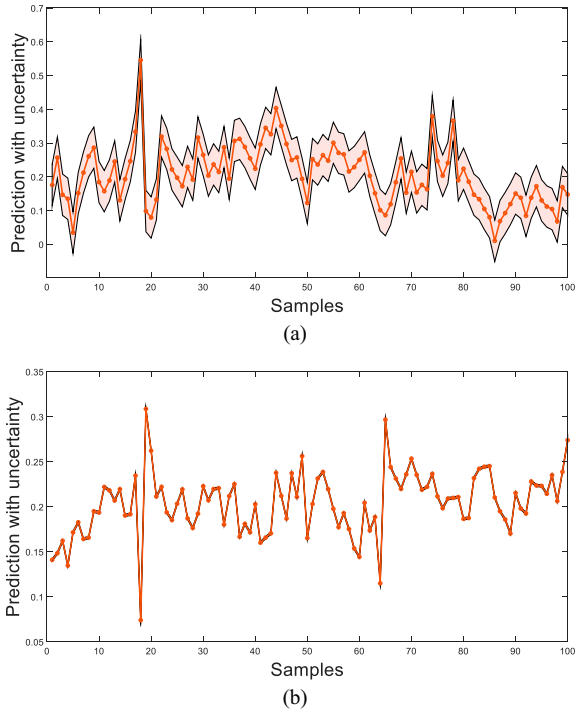


Fig. 9. Prediction results with uncertainties for the first 100 samples: (a) PPLS; and (b) deep PPLS.

layers of the deep model, the prediction uncertainty can be significantly reduced. Detailed results of the prediction uncertainty for the basic and the deep probabilistic PLS models for the first 100 testing data samples are shown in Fig. 9. As can be

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TABLE VII
PREDICTION VARIANCES AMONG DIFFERENT MODELS

Models	Layer #1	Layer #2	Layer #3	Layer #4	Layer #5	Deep PPLS
Prediction variance	0.0022	0.0023	0.0017	0.0028	0.0005	6.67×10^{-5}

TABLE VIII
RMSE VALUES UNDER DIFFERENT NUMBERS OF SELECTED LATENT VARIABLES

Number of Latent Variables	12	13	14	15	16
PPLS	0.135	0.1328	0.2545	0.4817	0.1608
Deep PPLS	0.1424	0.1442	0.159	0.1471	0.1454
Number of Latent Variables	17	18	19	20	
PPLS	0.1338	0.2836	1.0206	0.1624	
Deep PPLS	0.1426	0.1444	0.1497	0.1426	

seen, the prediction uncertainty of the deep model is very small compared to that of the basic shallow model. Similar results can be obtained by comparing different layerwise models to the final probabilistic model, which are shown in Table VII. Therefore, while the deep modeling structure largely increases the model capacity, the additional information fusion model can significantly improve prediction uncertainty for the deep model. Similarly, sensitivity analyses for parameters in the deep probabilistic PLS model are also provided. Table VIII shows the change of the RMSE for different numbers of selected latent variables in both the shallow and the deep probabilistic PLS models. It can be seen that the prediction performance of the basic PPLS model is quite sensitive to the number of selected latent variables. In contrast, the RMSE is relatively stable as a function of the change in the number of latent variables selected. For the number of hidden layers, the performance of the deep model does not seem to be quite sensitive to the change of the parameter, unless the number has been selected as a large value. In this case, however, the computational burden of the model will be significantly increased, which is not suitable in practice.

V. CONCLUSION

In this article, a deep residual form of the probabilistic PLS model has been proposed, which was inspired by the main idea of the recently developed lightweight deep latent variable model. Using deep learning on the simple latent variable model, both effectiveness and modeling efficiency can be obtained. Based on layerwise probabilistic modeling and extraction of supervised latent variables, an additional probabilistic PLS model has been constructed for information fusion and reextraction of supervised latent variables, which has the potential to reduce prediction uncertainty and thus keep the deep model more stable in practical applications. Two industrial examples have been provided for case studies of soft sensor application based on the developed method. Compared to the shallow probabilistic model, the performance of the deep model has been improved by 10%–20%.

Although better prediction performance has been validated, there are several critical issues that deserve further study. First, how to evaluate the importance and value of each extracted

$$
\begin{aligned}
&p\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right)\propto p\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right)p\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\right)\\
&\Downarrow\\
&\left[\begin{array}{l}
p\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix},\mathbf{P}^{(1)},\mathbf{Q}^{(1)},\mathbf{C}^{(1)},\boldsymbol{\Sigma}^{(1)}_x,\boldsymbol{\Sigma}^{(1)}_y\right)=\frac{1}{(2\pi)^{(m+r)/2}}\frac{1}{\left|\begin{bmatrix}\boldsymbol{\Sigma}^{(1)}_x & 0\\ 0 & \boldsymbol{\Sigma}^{(1)}_y\end{bmatrix}\right|^{1/2}}\\
\qquad\exp\left\{-\frac{1}{2}\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}-\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\right)^T\begin{bmatrix}\boldsymbol{\Sigma}^{(1)}_x & 0\\ 0 & \boldsymbol{\Sigma}^{(1)}_y\end{bmatrix}^{-1}\left(\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}-\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C} & \mathbf{0}\end{bmatrix}\right)\right\}\\
p\left(\mathbf{t}^s\right)=\frac{1}{(2\pi)^{q_s/2}}\exp\left\{-\frac{1}{2}\mathbf{t}^{sT}\mathbf{t}^s\right\},p\left(\mathbf{t}^u\right)=\frac{1}{(2\pi)^{q_u/2}}\exp\left\{-\frac{1}{2}\mathbf{t}^{uT}\mathbf{t}^u\right\}
\end{array}\right]\\
&\Downarrow\\
&E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right)=\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}^T\left(\begin{bmatrix}\boldsymbol{\Sigma}^{(1)}_x & 0\\ 0 & \boldsymbol{\Sigma}^{(1)}_y\end{bmatrix}+\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}^T\right)^{-1}\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}\\
&E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\mathbf{t}^{s(1)T}\\ \mathbf{t}^{u(1)}\mathbf{t}^{u(1)T}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right)=\mathbf{I}-\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}^T\left(\begin{bmatrix}\boldsymbol{\Sigma}^{(1)}_x & 0\\ 0 & \boldsymbol{\Sigma}^{(1)}_y\end{bmatrix}+\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}^T\right)^{-1}\begin{bmatrix}\mathbf{P}^{(1)} & \mathbf{Q}^{(1)}\\ \mathbf{C}^{(1)} & \mathbf{0}\end{bmatrix}\\
&\qquad+E\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right)E^T\left(\begin{bmatrix}\mathbf{t}^{s(1)}\\ \mathbf{t}^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix},\boldsymbol{\Theta}^{(1)}_{\text{old}}\right).
\end{aligned}
\tag{31}
$$

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latent variable in the hidden layers of the deep probabilistic model? Second, how to theoretically determine the number of supervised latent variables retained in each probabilistic PLS model? Third, if the number of hidden layers can be automatically determined, the uncertainty of the deep probabilistic model may be significantly reduced. A possible way is to connect these problems to the performance of the model, such as prediction accuracy, and model interpretability. Fourth, as can be seen from the results obtained from the two examples, there are still rooms to further improve the performance of the proposed method. From the aspect of model structure, it can be extended to dynamic or nonlinear forms. Finally, information fusion for different layers of the deep model can be further enhanced, which may significantly improve both prediction accuracy and robustness in more complicated stochastic modeling environments.

APPENDIX A
DETAILED DERIVATIONS OF THE EM ALGORITHM

As has been mentioned, instead of directly maximizing the log-likelihood function of the marginal distribution for each data sample, the EM algorithm tries to maximize the expected complete-data log-likelihood function given in (5). Here, the

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\boldsymbol{\Theta}^{(1)})]}{\partial \mathbf{P}^{(1)}}=0 \Rightarrow \sum_{i=1}^{n}\left[\begin{bmatrix}\boldsymbol{\Sigma}_x^{(1)} & 0\\ 0 & \boldsymbol{\Sigma}_y^{(1)}\end{bmatrix}^{-1}\left(\mathbf{x}_i-\mathbf{P}^{(1)}E\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)E^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]=0$$

$$\Downarrow$$

$$\mathbf{P}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{x}_iE^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(1)}\mathbf{t}_i^{s(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{32}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\boldsymbol{\Theta}^{(1)})]}{\partial \mathbf{Q}^{(1)}}=0 \Rightarrow \sum_{i=1}^{n}\left[\begin{bmatrix}\boldsymbol{\Sigma}_x^{(1)} & 0\\ 0 & \boldsymbol{\Sigma}_y^{(1)}\end{bmatrix}^{-1}\left(\mathbf{x}_i-\mathbf{Q}^{(1)}E\left(\mathbf{t}_i^{u(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)E^T\left(\mathbf{t}_i^{u(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]=0$$

$$\Downarrow$$

$$\mathbf{Q}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{x}_iE^T\left(\mathbf{t}_i^{u(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{u(1)}\mathbf{t}_i^{u(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{33}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\boldsymbol{\Theta}^{(1)})]}{\partial \mathbf{C}^{(1)}}=0 \Rightarrow \sum_{i=1}^{n}\left[\begin{bmatrix}\boldsymbol{\Sigma}_x^{(1)} & 0\\ 0 & \boldsymbol{\Sigma}_y^{(1)}\end{bmatrix}^{-1}\left(\mathbf{y}_i-\mathbf{C}^{(1)}E\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)E^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]=0$$

$$\Downarrow$$

$$\mathbf{C}^{(1)\text{new}}=\left[\sum_{i=1}^{n}\mathbf{y}_iE^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]\left[\sum_{i=1}^{n}E\left(\mathbf{t}_i^{s(1)}\mathbf{t}_i^{s(1)^T}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]^{-1} \tag{34}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\boldsymbol{\Theta}^{(1)})]}{\partial \boldsymbol{\Sigma}_x^{(1)}}=0 \Rightarrow \boldsymbol{\Sigma}_x^{(1)}\left[\mathbf{P}^{(1)\text{new}}\ \ \mathbf{Q}^{(1)\text{new}}\right]-\sum_{i=1}^{n}\left[\left(\mathbf{x}_i-\left[\mathbf{P}^{(1)\text{new}}\ \ \mathbf{Q}^{(1)\text{new}}\right]E\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right)E^T\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]=0$$

$$\Downarrow$$

$$\boldsymbol{\Sigma}_x^{(1)\text{new}}=\frac{1}{n}\text{diag}\left\{\sum_{\text{i}=1}^{\text{n}}\left[\mathbf{x}_\text{i}\mathbf{x}_\text{i}^\text{T}-\left[\mathbf{P}^{(1)\text{new}}\mathbf{Q}^{(1)\text{new}}\right]\text{E}\left(\begin{bmatrix}\mathbf{t}_i^{s(1)}\\ \mathbf{t}_i^{u(1)}\end{bmatrix}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{x}_\text{i}^\text{T}\right]\right\} \tag{35}$$

$$\frac{\partial E[L(\mathbf{X},\mathbf{Y},\boldsymbol{\Theta}^{(1)})]}{\partial \boldsymbol{\Sigma}_y^{(1)}}=0 \Rightarrow \boldsymbol{\Sigma}_y^{(1)}\mathbf{C}^{(1)\text{new}}-\sum_{i=1}^{n}\left[\left(\mathbf{y}_i-\mathbf{C}^{(1)\text{new}}E\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right)E^T\left(\mathbf{t}_i^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\right]=0$$

$$\Downarrow$$

$$\boldsymbol{\Sigma}_y^{(1)\text{new}}=\frac{1}{n}\text{diag}\left\{\sum_{\text{i}=1}^{\text{n}}\left[\mathbf{y}_\text{i}\mathbf{y}_\text{i}^\text{T}-\mathbf{C}^{(1)\text{new}}\text{E}\left(\mathbf{t}_\text{i}^{s(1)}\middle|\begin{bmatrix}\mathbf{x}_i\\ \mathbf{y}_i\end{bmatrix}\right)\mathbf{y}_\text{i}^\text{T}\right]\right\}. \tag{36}$$

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complete data includes the observed input and output variables $\mathbf{x}$, $\mathbf{y}$ and the latent variables $\mathbf{t}^s$, $\mathbf{t}^u$, and the latent variables are considered as missing data in the design of the EM algorithm. Based on the assumptions of a Gaussian distribution for both the latent variables and the measurement noises, the conditional probability distributions of input and output variables can be easily formulated, which are also Gaussian distributions.

In the E-step of the EM algorithm, the first and second sufficient statistics for the latent variables are required to determine the expected value of the complete-data log-likelihood function, and based on the Bayes' rule, the posterior probability of the latent variables can be calculated as

Then, in the M-step, we seek to update the model parameters by maximizing the expected complete-data log likelihood. To do this, the new parameters can be calculated by setting the partial derivative of the expected complete-data log likelihood function with respect to each parameter as zero. Differentiating and solving for the optimal points gives (32)–(36), shown at the bottom of the previous page.

More detailed information can be found in [40], [41], and [42].

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Zhiqiang Ge (Senior Member, IEEE) received the B.Eng. and Ph.D. degrees in automation from the Department of Control Science and Engineering, Zhejiang University, Hangzhou, China, in 2004 and 2009, respectively.

Currently, he is with the School of Mathematics, Southeast University, Nanjing, China. He was with the Department of Chemical and Biomolecular Engineering, Hong Kong University of Science Technology, Hong Kong, China; the Department of Control Science and Engineering, Zhejiang University; and Peng Cheng Laboratory, Shenzhen, China, from 2009 to 2024. His research interests include industrial big data, process monitoring, soft sensor, data-driven modeling, machine intelligence, and knowledge automation.

Dr. Ge was an Alexander von Humboldt Research Fellow with the University of Duisburg-Essen from 2014 to 2017, and also a JSPS Invitation Fellow with Kyoto University from June to August 2018.



Duxin Chen (Member, IEEE) received the B.S. degree in automatic control and the Ph.D. degree in control science and engineering from Huazhong University of Science and Technology, Wuhan, China, in 2013 and 2018, respectively.

Currently, he is an Associate Professor with the School of Mathematics and the Vice Director of the working committee of Jiangsu Provincial Scientific Research Center of Applied Mathematics, Southeast University, Nanjing, China. He is also one of the Direction Leaders of Huawei-SEU Joint Innovation Lab of Networked Collective Intelligence, Nanjing, China. His research interests include causal inference, prediction/generation, and system identification techniques for understanding complex networks and systems, and artificial intelligence related theory and applications.



Wenwu Yu (Senior Member, IEEE) received the B.Sc. degree in information and computing science and the M.Sc. degree in applied mathematics from the Department of Mathematics, Southeast University, Nanjing, China, in 2004 and 2007, respectively, and the Ph.D. degree in electronic engineering from the Department of Electronic Engineering, City University of Hong Kong, Hong Kong, China, in 2010.

Currently, he is the Dean of the School of Mathematics, the Deputy Director of Jiangsu Provincial Scientific Research Center of Applied Mathematics, and a Full Professor with the Endowed Chair Honor, Southeast University; and the Deputy Associate Director of the National Center of Applied Mathematics, Southeast University of Jiangsu, Nanjing, China. He held several visiting positions in Australia, China, Germany, Italy, The Netherlands, and USA. His research interests include multiagent systems, complex networks and systems, disturbance control, distributed optimization, machine learning, game theory, cyberspace security, smart grids, intelligent transportation systems, big-data analysis, etc.

Dr. Yu serves as an Editorial Board Member of several flag journals, including IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II, IEEE TRANSACTIONS ON INDUSTRIAL CYBER-PHYSICAL SYSTEMS, IEEE TRANSACTIONS ON FUZZY SYSTEMS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, *Science China Information Sciences*, *Science China Technological Sciences*, etc. He is the Editor-in-Chief of Complex Engineering Systems. He was listed by Clarivate Analytics/Thomson Reuters Highly Cited Researchers in Engineering from 2014 to 2024. He publishes about 100 IEEE Transactions journal papers with more than 20 000 citations. Moreover, he was a recipient of the Second Prize of the State Natural Science Award of China in 2016. He is also the Cheung Kong Scholars Programmer of the Ministry of Education of China (Artificial Intelligence).

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